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Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Does the sequence $\{a_n\}$ converge or diverge? Find the limit if the sequence is convergent.

$$a_n = \frac{4 - 8n^4}{n^4 + 9n^3}$$

Find the value of the limit. Note that ∞ does not represent a real number and that algebraic operations may not be performed on it. To find the given limit, first divide the numerator and denominator of the expression by the largest power of n that appears in either the numerator or denominator.

Divide both numerator and denominator of a_n by n^4 .

$$a_n = \frac{\frac{4 - 8n^4}{n^4}}{\frac{n^4 + 9n^3}{n^4}} = \frac{\frac{4}{n^4} - 8}{1 + \frac{9}{n}}$$

Substitute this expression for a_n into the limit.

$$\lim_{n \rightarrow \infty} \frac{4 - 8n^4}{n^4 + 9n^3} = \lim_{n \rightarrow \infty} \frac{\frac{4}{n^4} - 8}{1 + \frac{9}{n}}$$

Evaluate the following limits.

$$\lim_{n \rightarrow \infty} \frac{4}{n^4} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{9}{n} = 0$$

Using the information above, calculate the limit of the numerator.

$$\lim_{n \rightarrow \infty} \left(\frac{4}{n^4} - 8 \right) = -8$$

Also, calculate the limit of the denominator.

$$\lim_{n \rightarrow \infty} \left(1 + \frac{9}{n} \right) = 1$$

Finally, evaluate the limit of the entire expression.

$$\lim_{n \rightarrow \infty} \frac{4 - 8n^4}{n^4 + 9n^3} = \lim_{n \rightarrow \infty} \frac{\frac{4}{n^4} - 8}{1 + \frac{9}{n}} = \frac{-8}{1} = -8$$

Since $\lim_{n \rightarrow \infty} a_n = -8$, the sequence converges.